

A Topology of Variables Impacting the 2024 UK General Election Vote:

Additional Material

Appendix A: Data Pre-Processing & Exploratory Data Analysis

The main report used a merged dataset of Census 2021¹ and the 2024 UK General Election voting result²s across 575 constituencies in England and Wales to implement the correlation and BallMapper analysis. The Census 2021 datasets included were qualification levels (TS067), employment (TS065), income level through NS-SeC (TS062), migrant indicator (TS019), and household composition by age (TS003) – particularly selected in accordance to theoretical frameworks and empirical research findings on voting behaviour (Kulachai et al. 2023; McKnight & Cooper, 2022; McManus, 2023). The GE24 voting dataset on the other hand included information on GE24 and GE19, and whether the party holds, gains or loses their incumbency seats since the last elections. This is relevant information as the study sought to understand how voting behaviour has changed based on recent events.

Before merging, all dataset values were converted from object to float and only its percentage values (out of 100%) were taken instead of its raw values. These two were actioned for the purpose of standardisation to ease further coding analyses and interpretations. The list of percentage variables in the fully merged dataset is in Appendix A1.

¹Datasets can be accessed on here: https://www.nomisweb.co.uk/sources/census_2021_ts

²Voting results at constituency-level can be accessed through here <https://commonslibrary.parliament.uk/research-briefings/cbp-10009/>

A1. List of percentage variables in fully merged dataset

Variable	Description
level1_qualp level2_qualp level3_qualp level4_qualp no_qualp other_qualp	% where highest qualification is 1 % where highest qualification is 2 % where highest qualification is 3 % where highest qualification is 4 % where highest qualification is 0 % where highest qualification is Other
dep0p dep1p dep2p dep3p dep4p	% of households with 0 deprivation indicator % of households with 1 deprivation indicator % of households with 2 deprivation indicators % of households with 3 deprivation indicators % of households with 4 deprivation indicators
L1-L3p L4-L6p L7p L8-L9p L10-L11p L12p L13p xworkedunemployedp studentp	% of people with higher managerial and professional occupations % of people with lower managerial and professional occupations % of people with intermediate occupations % of small employers and own account workers % of people with lower supervisory and technical occupations % of people with semi-routine occupations % of people with routine occupations % of people never worked/unemployed long-term % of full-time students
mignotUKp migStup migUKp nonmigp	% of migrants from outside the UK % of migrants are students/boarding schools in the UK % of migrants from within the UK % of non-migrants (same address)
onep16-49p onep50-99p othermixed16-49p othermixed50-99p singlefam0-15p71 singlefam16-49p singlefam50-99p	% of 1-person households aged 16-49 % of 1-person households aged 50 above % of other household types (students/with dependent children) aged 16-49 % of other household types (students/with dependent children) aged 50 above % of single-family household types aged 0-15 % of single-family household types aged 16-49 % of single-family household types aged 50 above

Appendix A2 reports the descriptive statistics of each constituency on average, providing in-depth understanding of data across constituencies on average. Across the fully merged dataset, some key highlights include the percentage of valid voters which was

59.84% - this was considered significantly lower compared to GE19 (67.30%)³. This implies that inferences made from voting behaviour may not be fully representative of the population as it only accounts for a little more than half of each constituency on average, on top of the variable dataset being from Census 2021 collected during COVID-19 restrictions. In addition, it can be observed that variables within a category have very different ranges of values. For example, whilst the deprivation on all 4 indicators ranged between 0-0.8%, those having no deprivation indicators within a constituency ranged between 32.1% and 62.7%. These can disable accurate comparisons between variables. Therefore all variables are normalised [0, 1] to mitigate difference in means and ranges that may skew evaluations, typically done in voting studies employing TDABM like Otway & Rudkin (2024). In light of these limitations, overall, the descriptives found alignment with current literature reported in the main report. Furthermore, to understand how recent vote shares differed this time around, Appendix A3 reports on the exploratory data analysis for the GE19 vs GE24 vote shares.

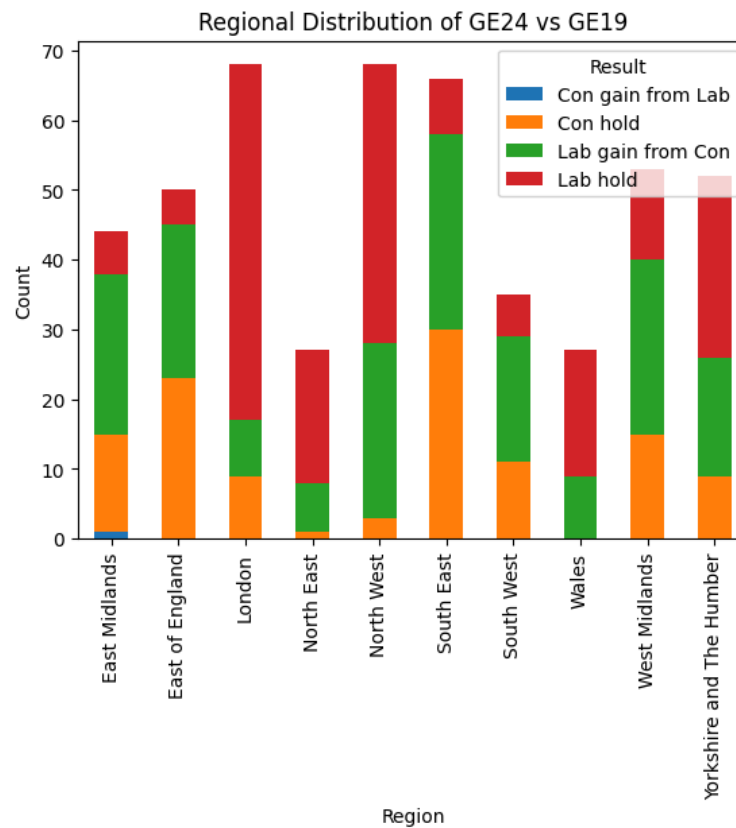
³ Voting results for GE19 can be accessed through this link:
<https://commonslibrary.parliament.uk/research-briefings/cbp-8749/>

A2. Full merged dataset Summary Statistics for the GE24 vote and its variables across 575 constituencies (%)

% for 575 Constituencies	Variable	mean	std	min	max
2024 GE Vote	Valid Voters	59.84	7.04	39.99	75.21
	Conservative	24.73	10.12	0	53.3
	Labour	35.51	13.86	0	70.6
	Lib Dem	11.97	13.13	0	62.7
	Other	27.42	12.16	6.3	174.3
Qualifications (education)	Level 1	10.24	2.02	3	14.6
	Level 2	14.29	2.46	6.2	18.6
	Level 3	17.98	2.90	10.4	36.6
	Level 4	35.39	9.02	19.5	66.4
	None	19.21	4.88	4.9	34.2
	Other	2.89	0.49	1.4	5.4
Deprivation Indicator (income level)	None	48.41	5.94	32.1	62.7
	1	33.46	1.85	27.1	38.9
	2	14.20	3.11	7.5	22.4
	3	3.70	1.51	1.2	8.3
	4	0.23	0.13	0	0.8
NS-SeC (employment)	L1-L3	10.69	4.13	3.7	25.9
	L4-L6	16.36	2.71	8.2	21.9
	L7	9.39	1.37	4.9	13.3
	L8-L9	8.68	2.15	4	15.7
	L10-L11	4.44	1.03	1.7	7.8
	L12	9.32	1.96	4	14.1
	L13	9.82	2.99	3.1	19.9
	Never worked/Unemployed	6.81	2.38	3.1	16.3
	Full-time Student	6.10	4.13	2.9	38.7
Migrant Indicator	International	0.87	0.77	0.14	6.32
	Domestic	9.32	2.79	5.69	28.71
	Student	0.57	1.07	0.07	9.5
	Non-migrant	88.23	4.31	56.89	92.85
Household Composition (by age)	One-person (16-49)	3.71	1.57	1.5	13.3
	One-person (>50)	9.15	1.73	3.3	14.3
	Other/Mixed (16-49)	7.04	4.95	0	34
	Other/Mixed (>50)	3.17	0.99	0	8.9
	Single-family (<15)	16.88	1.97	9.8	22.2
	Single-family (16-49)	31.99	2.98	18	45.1
	Single-family (>50)	26.28	6.45	8.3	40.6

Note: Parties with lower proportions of votes (e.g. independent, Reform UK, Green Party) are combined into a category “Other”. Age ranges for household composition was grouped by under 50/over 50 instead of the original dataset age range and considered demographics above 16 as the minimum voting age.

A3. Exploratory Data Analysis of 2024 vs 2019 vote shares



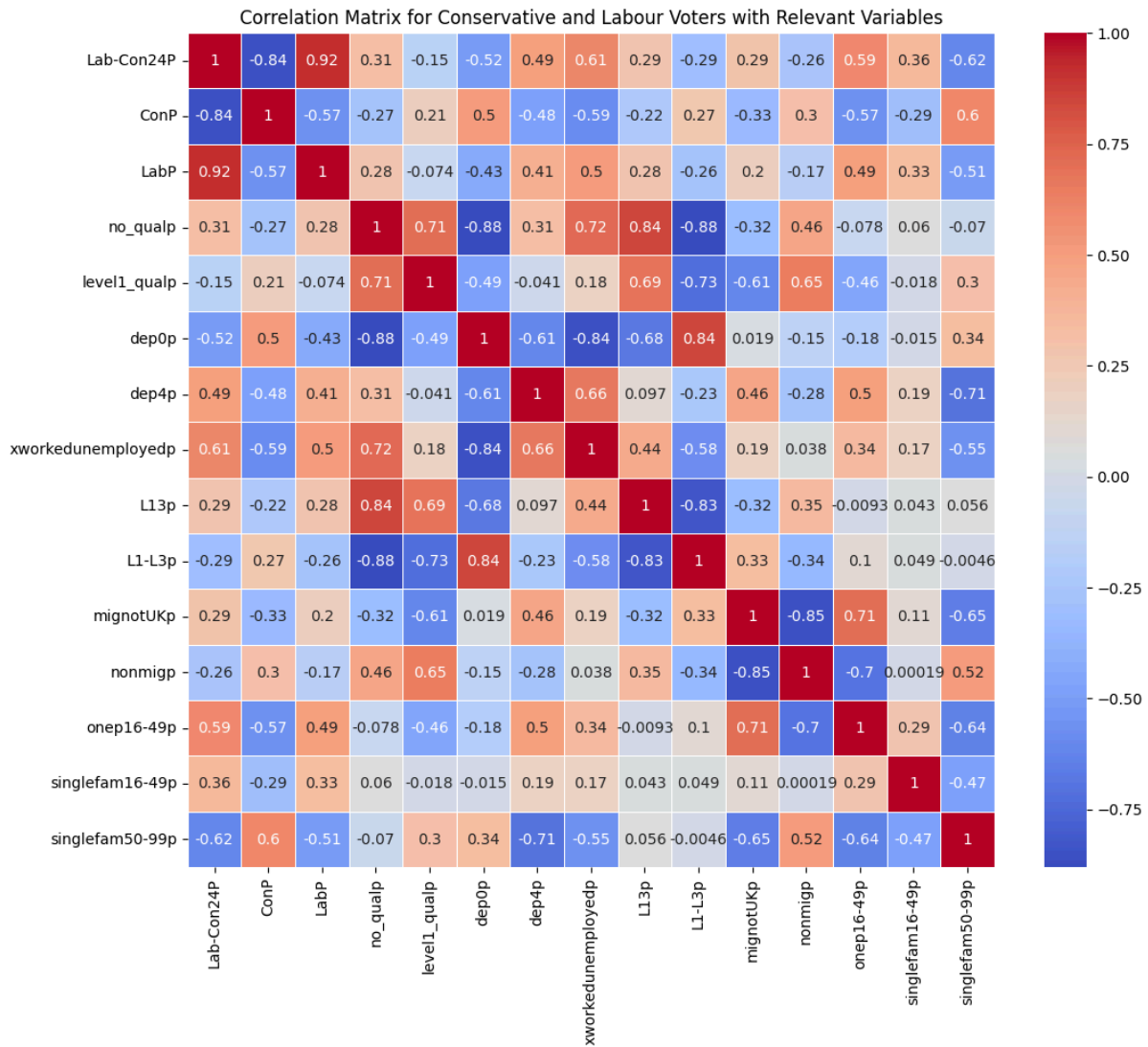
In comparing GE24 with GE19 across all 575 constituencies, Conservative only gained from Labour in one constituency in GE24 – Leicester East (East Midlands). Despite this win, the Tories still had 7.4% less votes compared to GE19. A stark difference between the two elections is also Labour gaining 182 seats from previously Conservative areas, suggesting a potential regain in Labour Red Wall areas as previously hypothesised, particularly in North West and Yorkshire regions (English, 2022; Dijkstra et al., 2020). Interestingly, the 5 seats won by Reform UK in GE24 were all Conservative seats⁴. Referring back to pocketbook voting (Lewis-Beck, 1985), these findings provide evidence that due to recent economic crises voters have switched incumbent party support based on their self-interest.

⁴ Analysis comparing GE24 and GE19 can be referred to in the code below. The finding of Reform UK seats being previously Conservative was obtained from the GE 24 voting result dataset.

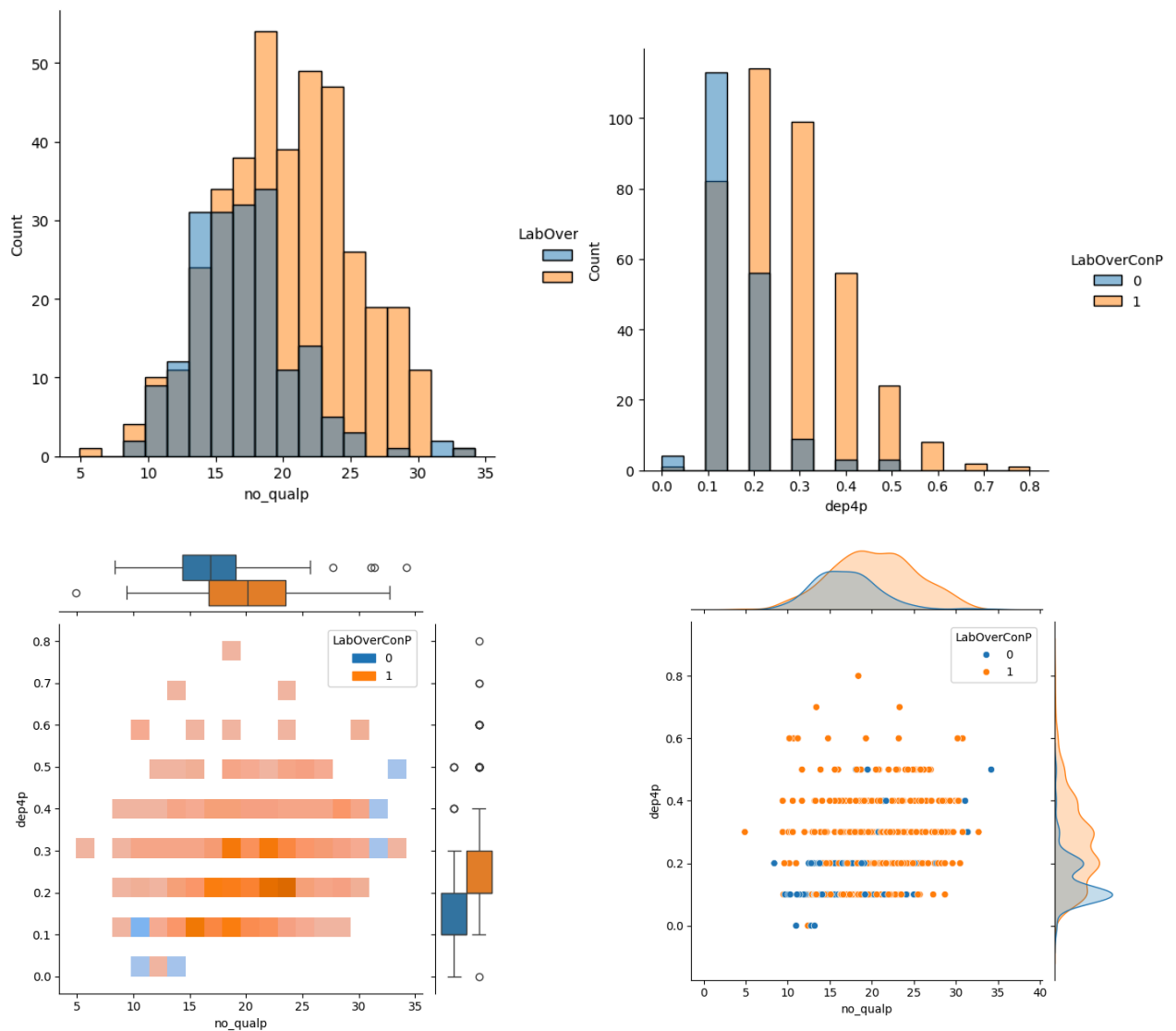
Since 2019 GE, Labour strongholds have been in regions of London, North West, and North East (gains and holds). Most seats were maintained in London while the Tories held the most in the South East. The former's inclination for Labour explains Kulachai et al. (2023)'s finding that urban areas favoured more left-wing policies. Conservatives seem to mostly hold in East Midlands, East of England, South East, with one seat gain from Labour in the East Midlands. These regions are known to be traditionalist voters in more rural areas (Kanagasooriam, 2019; Curtice, 2021). Wales, West Midlands, and Yorkshire show relatively even distribution between parties - thus not showing obvious party polarisation changes within the 5 years. These findings aligned with literature on Red Wall and Blue wall areas indicating that the strongholds through past elections have maintained to an extent.

Next, a correlation analysis was carried out, visualised through a heatmap to view the statistical relationships between variables as our literature review reported. All linear relationships aligned with empirical findings, but variables very highly correlated ($r > 0.7$) were not utilised for analysis as it may suggest potential multicollinearity. The heatmaps were also used to filter out any relevant/irrelevant variables picked for analysis, as regression approaches like Ordinary Least Squares can only be inferred through limited variables, notwithstanding TDABM being a tool ideal for multidimensional variables. The exploratory data analysis carried out to study bivariate relationships between variables are observed in Appendix A4 and A5.

A4. Correlation Matrix Heatmap



A5. Distribution Plots

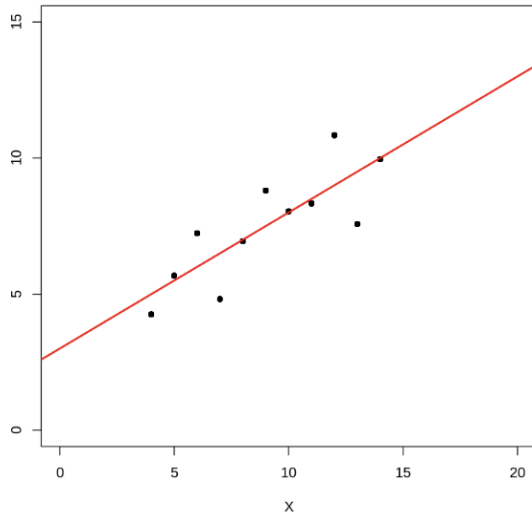


1. In this histogram, the orange bar indicates areas where Labour won over Conservatives, while the blue bar represents constituencies where Conservatives had better results. It further demonstrates how Labour performed better in areas constituting voters who had not worked or were unemployed, aligning with the trends observed prior to this.

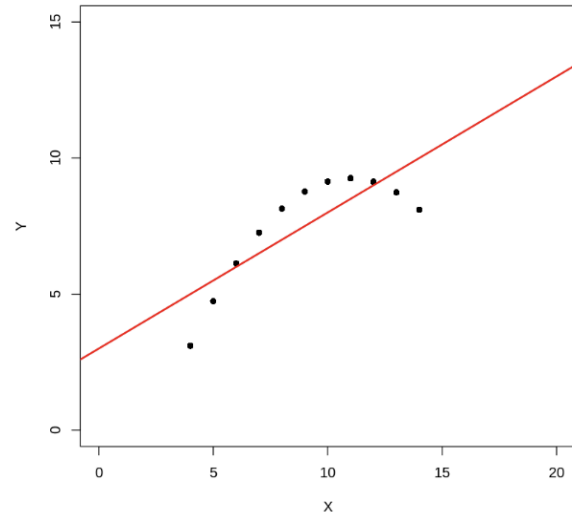
2. In the histogram on non-migrant proportions and voting choice, Labour dominated areas observed a slightly lower percentage of non-migrant voters compared to the contrary. Labour also performed better in more diverse areas with more migrant voters, while Conservative areas were more homogeneous in their votes with higher non-migrants.
3. The hexbin and boxplots between non-migrant voters and those who are unemployed also agrees with scatterplot visualisations suggesting denser proportions in Conservative-dominant areas where unemployment rates are lower, and non-migrant population rates higher.
4. Joint plot of scatterplot with kernel density estimators suggest that in Labour dominated constituencies, there is a slight positive relationship where the higher the local populations, the higher the proportions of unemployed. Though this is more spread out than the localised Conservative areas. KDE shows Labour constituencies are more common with higher unemployment and more diverse populations.

Appendix B: Ordinary Least Squares section

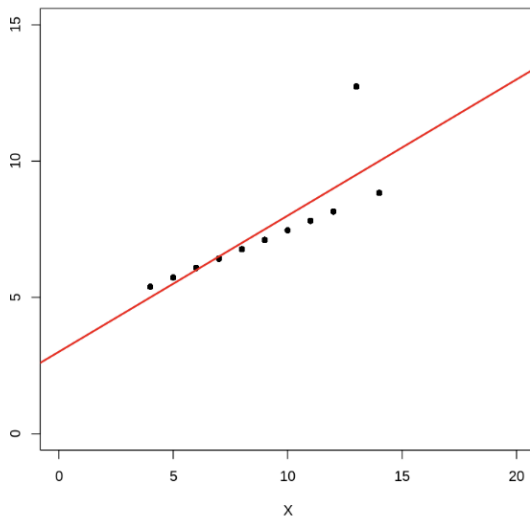
B1. Anscombe (1973)



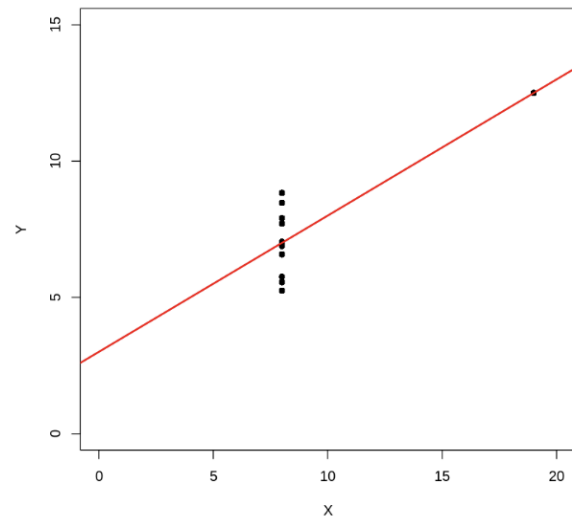
(a)



(b)



(c)



(d)

(a) Noisy, (b) Non-linear, (c), outlier, (d) high leverage

Anscombe (1973)'s four datasets of linear regression in Appendix B1 shows how all four models have similar goodness-of-fit statistics despite having distinct data structures. This underscores that using models with an outcome variable does not necessarily show the shape of joint distributions, making appropriate visualisation of models important to show accurate relationships. Nevertheless, regression models still serve its advantage of understanding significance of relationships between variables.

With that being said, for this study we maintain this use, complemented by TDABM to offer deeper comprehension of joint distributions. We take Anscomb's findings into consideration by also visualising scatterplots from linear associations of variables - these were reported in the main study and further compared with TDABM. In implementing the OLS model, the fully merged dataset was reduced as only this way would multicollinearity and overfitting be mitigated in this traditional approach. Descriptive statistics and the heatmaps of this reduced dataset are shown in Appendix B2 and Appendix B3 respectively to filter out irrelevant variables from the previous summary statistics.

B2. Reduced dataset summary statistics

	Lab-Con24P	no_qualp	dep4p	xworkedunemployedp	nonmigp	singlefam16-49p
count	575.0	575.0	575.0	575.0	575.0	575.0
mean	10.8	19.2	0.2	6.8	88.2	32.0
std	21.3	4.9	0.1	2.4	4.3	3.0
min	-37.6	4.9	0.0	3.1	56.9	18.0
25%	-5.9	15.6	0.1	5.0	87.9	30.3
50%	10.7	18.8	0.2	6.2	89.6	32.4
75%	26.6	22.6	0.3	8.3	90.7	33.8
max	67.1	34.2	0.8	16.3	92.8	45.1

B3. Correlation Matrix Heatmap of Reduced Dataset



As mentioned, the filtered dataset excludes any multicollinearity ($r > 0.7$) observed in the above heatmaps. Correlation revealed the strongest relationships with the Labour vote being those unemployed or not working for a long time, followed by voters who were deprived at all levels, younger populations in single-family households, have no qualification and of non-migrant status, aligning with literature reported in the main report.

B4. OLS Regression Model (non-robust)

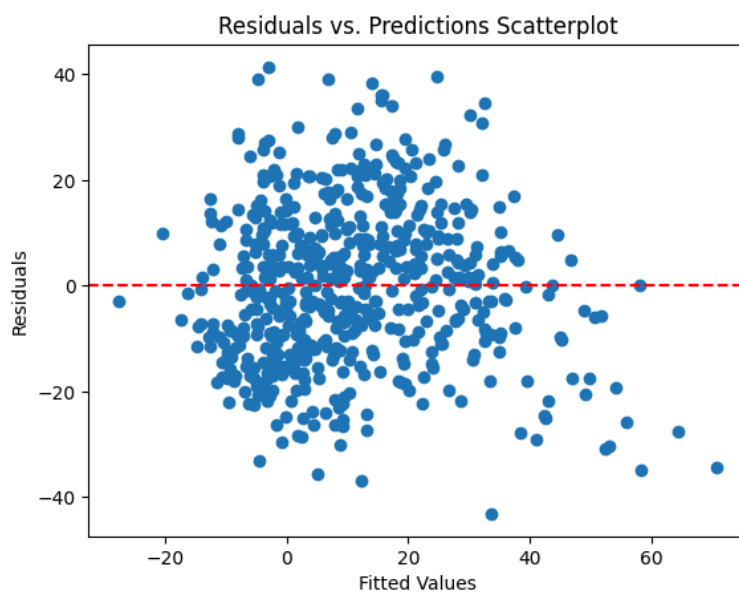
OLS Regression Results						
Dep. Variable:	Lab-Con24P	R-squared:	0.511			
Model:	OLS	Adj. R-squared:	0.506			
Method:	Least Squares	F-statistic:	118.8			
Date:	Fri, 13 Sep 2024	Prob (F-statistic):	6.49e-86			
Time:	17:02:29	Log-Likelihood:	-2368.5			
No. Observations:	575	AIC:	4749.			
Df Residuals:	569	BIC:	4775.			
Df Model:	5					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	43.9721	16.714	2.631	0.009	11.143	76.802
no_qualp	0.1325	0.238	0.558	0.577	-0.334	0.599
dep4p	-2.2737	6.990	-0.325	0.745	-16.003	11.456
xworkedunemployedp	5.0234	0.517	9.721	0.000	4.008	6.038
nonmigp	-1.4567	0.194	-7.518	0.000	-1.837	-1.076
singlefam16-49p	1.8465	0.215	8.585	0.000	1.424	2.269
Omnibus:	3.593	Durbin-Watson:	1.775			
Prob(Omnibus):	0.166	Jarque-Bera (JB):	3.449			
Skew:	0.143	Prob(JB):	0.178			
Kurtosis:	2.750	Cond. No.	2.59e+03			
Notes:						
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.						
[2] The condition number is large, 2.59e+03. This might indicate that there are strong multicollinearity or other numerical problems.						

With reference to the main report, Appendix B4 shows the OLS regression model before application of robust standard errors. This was applied to address the heteroscedasticity found through the assumption checks shown in Appendix B5.

B5. Assumption Checks

	0
const	717.6
no_qualp	3.4
dep4p	2.1
xworkedunemployedp	3.9
nonmigp	1.8
singlefam16-49p	1.1
dtype: float64	

Note: VIF results show that multicollinearity is not a serious issue in the model, except for the constant term being above 10 - which is typical.



Note: In terms of homoscedasticity- plots are somewhat scattered, with very mild heteroscedasticity around fitted values near 40. This potential non-linearity highlights the need to use a non-linear model to visualise relationships. We continue conducting a homoscedasticity and heteroscedasticity test for further diagnostics.

```
[ ] # Homoscedasticity - Breusch-Pagan test
from statsmodels.stats.diagnostic import het_breuschpagan
bp_test = het_breuschpagan(results.resid, results.model.exog)
labels = ['LM Statistic', 'LM-Test p-value', 'F-Statistic', 'F-Test p-value']
print(dict(zip(labels, bp_test)))

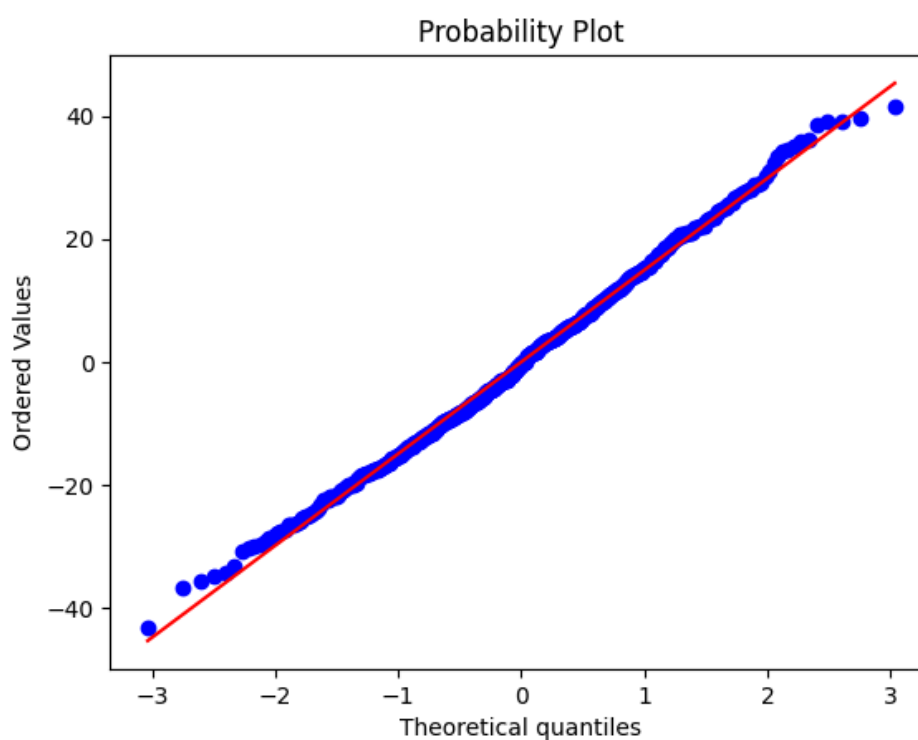
⇒ {'LM Statistic': 36.91358469989023, 'LM-Test p-value': 6.233196124383069e-07, 'F-Statistic':
```

```
[ ] # Heteroscedasticity - Goldfeld-Quandt test

from statsmodels.stats.diagnostic import het_goldfeldquandt
gq_test = het_goldfeldquandt(residuals, results.model.exog)
print(f'Goldfeld-Quandt test: {gq_test}')
```

```
⇒ Goldfeld-Quandt test: (0.9330542975266694, 0.7192553482898352, 'increasing')
```

Note: For Breusch-Pagan test, $p\text{-value} < 0.05$ implies that we reject the null hypothesis - meaning there is heteroscedasticity. For Goldfeld-Quandt test, values close to 1 (0.933) suggest variances in both are similar. $p\text{-value} > 0.05$ implies that we do not reject the null hypothesis, therefore no significant evidence of heteroscedasticity. Overall, both tests suggest different assumptions, though the slight deviations in q-q plot tails below explain this heterogeneity.



The updated OLS regression model in Appendix B6 above corrects heteroscedasticity for more reliable statistical inferences. It adjusts the standard errors to reflect true variability without affecting the goodness of fit. Overall, the robust OLS model confirms that Labour tends to perform better in constituencies with younger, economically vulnerable populations. Additionally, the lack of significance for the deprivation variables suggests that economic hardship (dep4) alone does not fully explain voting patterns between Labour and Conservative constituencies. This fits with research showing that, while economic prosperity (or a lack thereof) are important, other socioeconomic variables become more prominent in driving voting behaviour, especially post-Brexit.

Appendix C: BallMapper Analysis

Since TDABM does not worry about correlation, we include other axis variables from the rest of the dataframe in our BallMapper analysis - selected based on literature. Balls or nodes represent clusters of constituencies with consideration of other axis socioeconomic, demographic and migration variables. Its sizes indicate the data points that lie within each ball, where larger balls suggest more points and smaller balls indicate the opposite or potential outliers. Ball colours represent the range of values of the axis variables. Edges indicate the overlap or relationships between constituencies. The role of seed is to initialise the randomness of TDABM plots/position and for reproducibility when comparing. The role of radius eps determines the distance of balls from each other, thus controlling the granularity and structure of TDABM outcomes.

Findings of analysis from this section can be referred to in the main report. In the following sub-appendices, information about ball memberships are expanded with reference to evaluations made in the main report. The full list of ball membership was saved into a csv-file, but the summary statistics are reported in Appendix C1 below. It is important to note that TDABM randomly chooses landmarks for balls, thus it is possible to get different ball memberships even with the use of the same radius. Rudkin et al. (2023) and Otway & Rudkin (2024) have conducted 1000 repetitions of their BM algorithm within a range of radii and assessed for its robustness, therefore we attest to the robustness of ball memberships with our chosen radii in this particular study instead of conducting own repetitions.

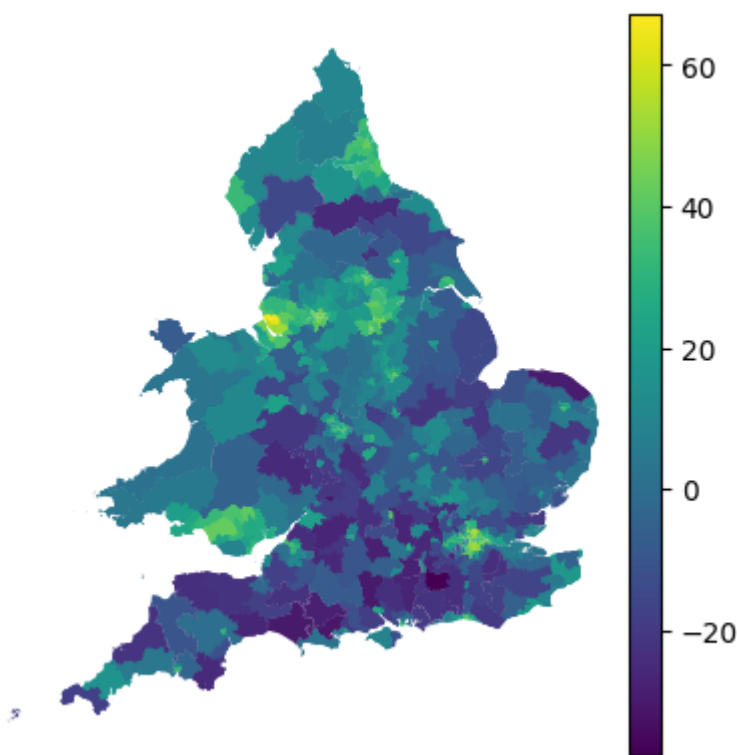
In Appendix C1, the values of each variable column represent its average value for each ball given the normalised data. It confirms with evaluations made from BM visualisations in the report, where Labour performed better in certain areas like ball 27 (51.4%) and ball 20 (50.1%), and Conservative strongly anchored in balls 26 and 17 (-15.3% and -4.0% respectively). To visualise the outcome variable of Labour over Conservative constituencies as well as ball membership respectively, the England and Wales map is shown in Appendix C2 below⁵.

⁵ The boundary map used for this dataset can be accessed through this link: <https://observablehq.com/@jwolondon/uk-election-2024-boundary-data>, while the csv file listing all the constituencies and its code commensurate to this map can be found in the GitHub repository linked at the end of this file.

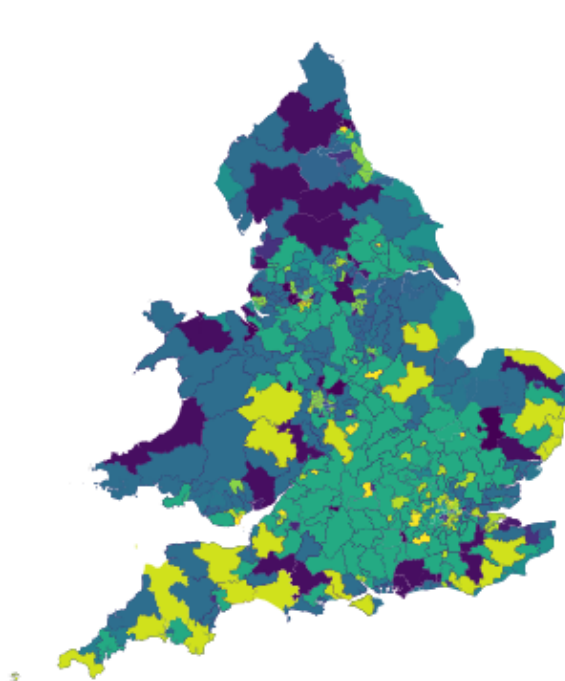
C1. Summary statistics of Ball membership

ball	Lab-Con24P	no_qualp	dep4p	xworkedunemployedp	nonmigp	singlefam16-49p
0	6.9	0.5	0.2	2.1e-01	0.9	0.5
1	-5.6	0.4	0.2	1.4e-01	0.9	0.5
2	24.4	0.7	0.4	4.3e-01	0.9	0.6
3	19.5	0.3	0.3	1.4e-01	0.6	0.4
4	31.3	0.3	0.4	2.7e-01	0.6	0.5
5	37.3	0.5	0.6	6.0e-01	0.6	0.5
6	26.1	0.8	0.5	8.5e-01	0.9	0.5
7	21.8	0.8	0.6	7.4e-01	1.0	0.5
8	36.4	0.7	0.6	9.8e-01	0.7	0.5
9	18.6	0.6	0.2	3.9e-01	0.9	0.5
10	6.8	0.5	0.3	2.6e-01	0.9	0.5
11	13.8	0.8	0.3	4.1e-01	0.9	0.5
12	22.6	0.4	0.5	2.9e-01	0.8	0.5
13	30.4	0.6	0.6	5.7e-01	0.9	0.5
14	21.2	0.6	0.4	4.8e-01	0.9	0.5
15	28.0	0.0	0.4	7.6e-03	0.2	0.4
16	23.9	0.2	0.8	3.9e-01	0.6	0.5
17	-4.0	0.3	0.2	1.2e-01	0.9	0.5
18	43.1	0.2	0.4	1.9e-01	0.0	0.3
19	36.0	0.5	0.4	4.3e-01	0.7	0.6
20	50.1	0.5	0.5	5.5e-01	0.5	0.4
21	45.2	0.5	0.9	5.7e-01	0.8	0.5
22	44.7	0.2	0.4	2.5e-01	0.2	0.6
23	32.2	0.8	0.4	6.1e-01	0.9	0.6
24	42.0	0.5	0.5	5.2e-01	0.8	1.0
25	27.2	0.4	0.4	3.2e-01	0.8	0.6
26	-15.3	0.5	0.2	1.6e-01	0.9	0.3
27	51.4	0.6	0.8	5.4e-01	0.8	0.6
28	14.8	0.3	0.2	1.4e-01	0.6	0.6

C2. Mapping Constituencies



Note: Plots are coloured based on *Labour over Conservative votes* in GE24. Negative values indicating Conservative winning over Labour can be seen more in the Southern regions, whilst Labour regions were more scattered.



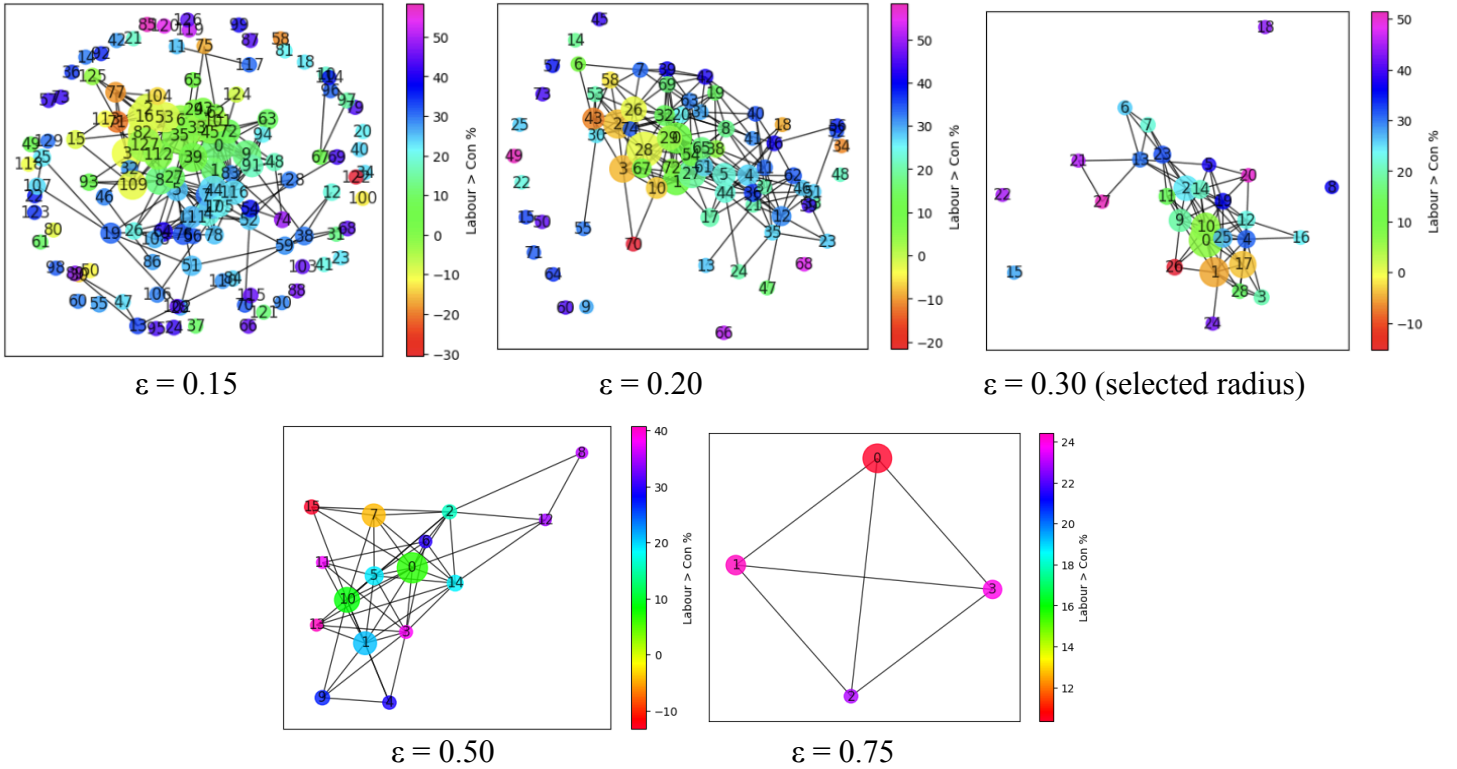
Note: Plots are coloured based on *ball membership* – there were a total of 28 balls for radius 0.3. Ball 0 had the highest frequency of shared constituencies, and dispersed. Plots agree with the urban-rural divide, where high membership regions tend to be urban constituencies, and middle England balls being more

homogenous and Conservative areas.

Appendix D: Robustness Checks

Robustness checks were conducted to verify the reliability of findings, as typically demonstrated in other similar voting studies of TDABM. This is done by adjusting key parameters. For this study, two robustness checks were carried out by 1) using different radii, and 2) using expanded axis variables for TDABM. The purpose of choosing these two was to check for any significant shifts in constituency clustering and to evaluate whether relationships were consistent when accounting for other levels of each variable (like dep0-3 for dep4). Ideally, conducting 1000 repetitions could attest for analysis robustness as demonstrated by Rudkin et al. (2023), but we infer their findings onto this similar study. Appendix D1 evaluates the robustness checks.

D1. Effect of changing radius for the Labour Over Constituency %

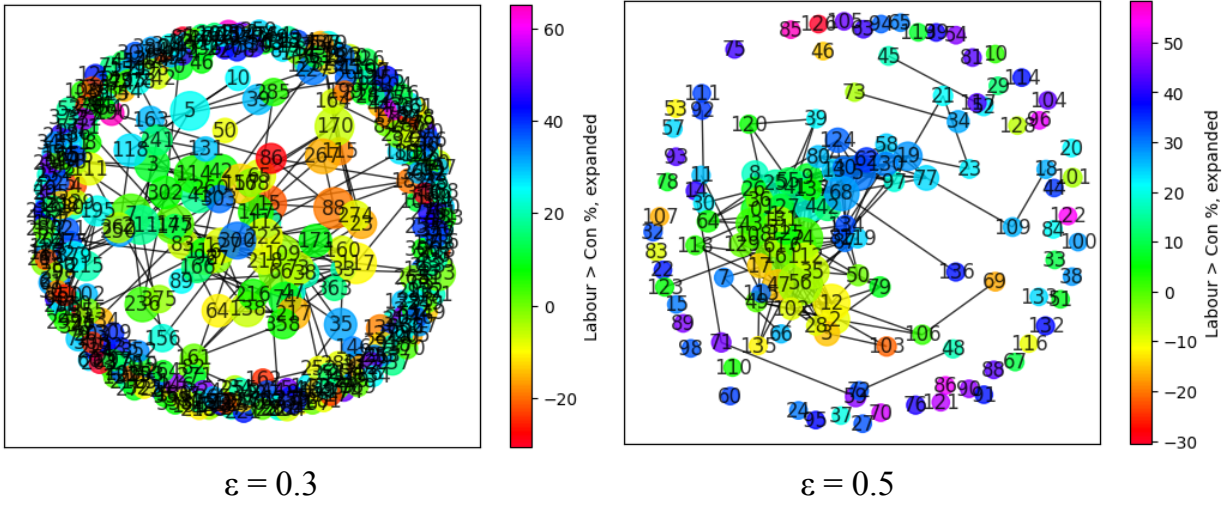


Appendix D1 demonstrated how as the radius increases, the balls observed are reduced, omitting the outliers by $\varepsilon = 0.50$. Evidently, outlying balls provide important context – radius selection beyond this is considered non-optimal as oversimplification loses its detail. Opposite to that, balls smaller than the default $\varepsilon = 0.3$ still showed consistent findings in the main clusters, but with excessive noise/disenjoinments.

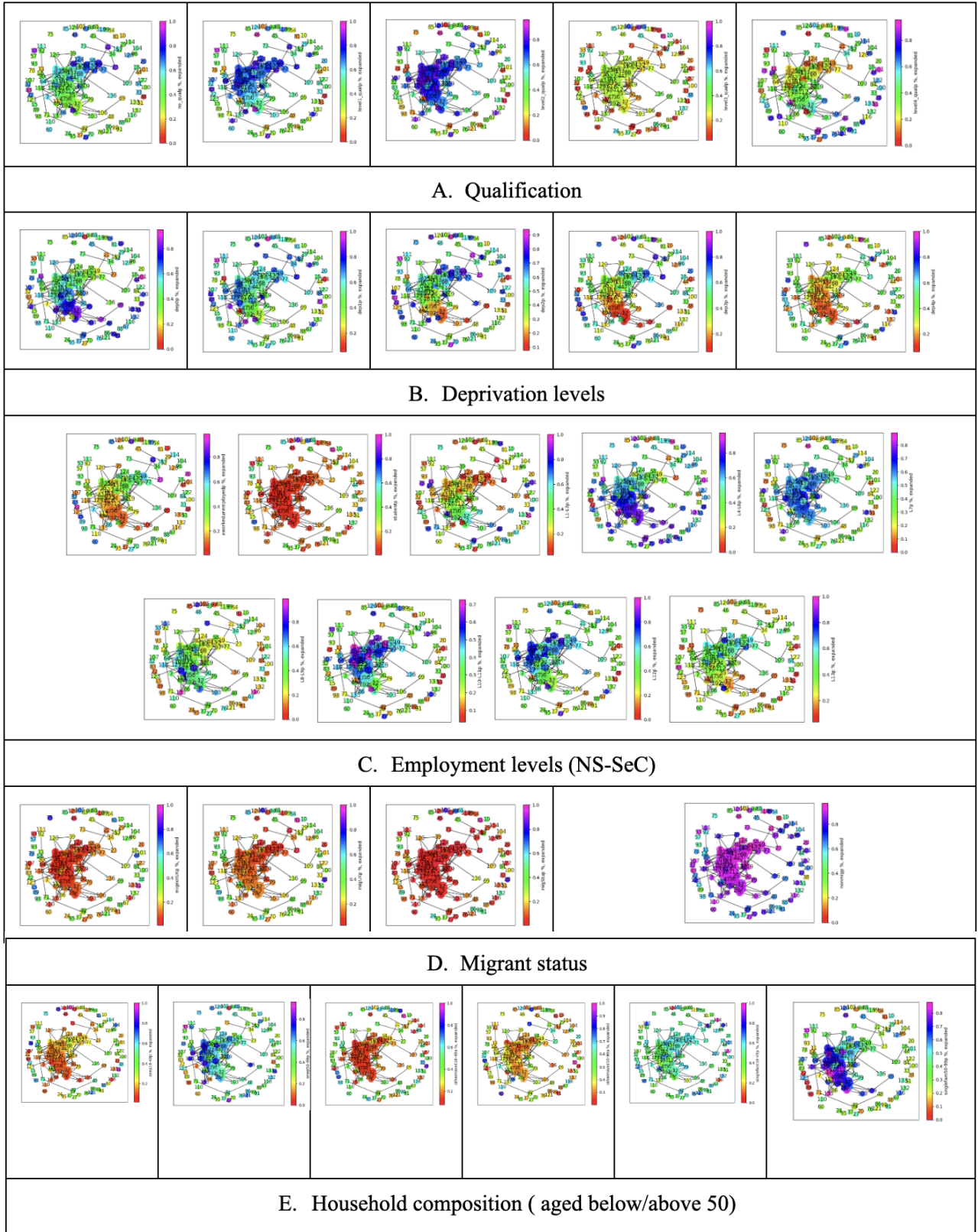
The different radii utilised did not significantly shift key findings, as the main clusters, though different landmarks, showed similar behaviours between shared constituencies in light of granularity being lost with further radius expansion. This suggests robustness to the radius choice. This is not unique as other studies conducting robustness checks on radius choice with even deeper analysis demonstrated consistent TDABM representations, especially considering the trade-off.

Since the purpose of visualisations in this study were to capture multidimensional relationships/patterns that impact voting behaviour, other radii choice risk harder interpretations and lesser generalisability of findings. Ideally, the selection of ε in BM graphs should be considered by a trade-off between preserving detailed nodes and lesser nodes for uncomplicated representations (Rudkin et al., 2023). Therefore, $\varepsilon = 0.3$ is deemed optimal because, 1) joint distributions consistently maintained with OLS correlations with further nuance (Dlotko et al., 2022), 2) this showed heterogeneity in Labour votes as opposed to Conservative areas in joint distributions, as Otway & Rudkin (2024)'s TDABM estimated, and 3) the outliers captured with this level of granularity eased interpretations made, with findings from this radius remaining with literature.

D2. Expanding axis variables



TDABM is advantageous over other visualisations when considering multiple axes. Therefore, we consider all variables from the dataset instead of limiting it as previously done when comparing with scatterplots, ensuring that the limited variables were still robust despite other dimensions missed. As seen in the Appendix above, using $\epsilon = 0.3$ provides very high granularity. Although this maximises context, its dense structure loses its ability to interpret visualisations accordingly. Therefore, to evaluate robustness on expanding axis variables, we increase the radius to $\epsilon = 0.5$ to provide a digestible visualisation sufficient enough for interpretation. The BallMapper results for each variable is shown below. Observed in this second assessment, incorporating more socioeconomic, demographic and migration variables generally offer further granularity, specifically in detecting smaller, more distinct clusters of Labour over Conservative-winning constituencies that were not as stark in the original analysis. Generally, robustness check of variable expansion remains consistent with original findings. For example, comparing the Labour over Conservative BallMapper plots above with the original BM plots evidently remains the clear distinctions of heterogeneity among Labour areas and homogeneity among the Conservatives, with richer colour intensity indicating deeper nuance of interpretations.



In light of voting behaviour being consistent with axis variable expansion, granularity of inferences and the smoothness of colour transitions based on the more detailed subgroups highlight how different regions differ based on specific attributes. For instance, for qualification variables (Appendix D2.A above) the main clusters showed different colour intensity for each level of qualification notwithstanding the colour homogeneity. Those with level 2 qualifications had the coolest colour intensity in the main cluster compared to the original finding of voters of no qualification making up moderate proportions in the joint distributions, implying the nuances for each subcategory of variables.

Whilst the main cluster from the original analysis showed moderate proportions of no qualification, unemployment, young voters in single-family households who were mostly non-migrant populations with low deprivation at all 4 levels, main clusters in the expanded model showed deeper insights, where balls mostly had critically low student populations and more lower-managerial/routine occupations. This helps explain the significantly low migration levels in the main cluster from the original analysis, as Jenner & Stockings (2017) explained how rural middle England areas, or the ‘left-behind’, show socioeconomic behaviours of low migration and routine occupations. Overall, this second check shows that the model’s use of limited variables due to regression was robust. However, would be more ideal in interpretations when including expanded variables.

Further studies should build on this model by adjusting parameters of using as many variables considered through empirical findings and theoretical frameworks. Evidently, considering multidimensions provide a more inclusive and comprehensive evaluation to joint distributions

Appendix E: Code

The code for the whole analysis can be accessed through the GitHub repository found in this link:

[https://github.com/amandaazman/voting_TDA/blob/main/erp%20\(2\).ipynb](https://github.com/amandaazman/voting_TDA/blob/main/erp%20(2).ipynb)

The repository also includes csv files of cleaned datasets for each socioeconomic, migration, and demographic dataset as well as the boundary map list for reference.